Strain gauge Work Instruction

Installation Guide

1.1. Required equipment

Tools:

Glass plate
5.5" scissors
Hb pencil
Ball point pen
Transparent adhesive tape
Fiber tip pen
Pocket size multimeter
Tie wrap cutter (side cutter)
Normal ruler
Screwdriver flat
Screwdriver Philips
Scalpel
Tweezer small
Flux pen
Soldering iron
Dremel multi tool

Consumables:

Solder
1 m Rainbow Cable
Heat shrink
Cleaning pads
RMS 1 spray
Teflon foil
Adhesive
Covering material 1

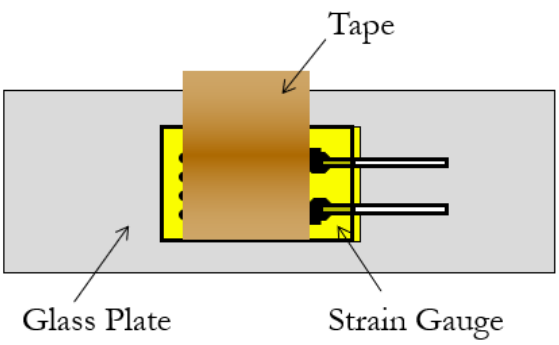
Covering material 2
Strain gauge
Self-adhesive cable tie saddles
Dremel 80 grit flapper wheel
220 grit sandpaper

1.2. Preparing the Strain Gauge

- Clean the glass plate with RMS 1 and cleaning pads
- Place the strain gauge on the glass plate using clean tweezers.

NB: Never handle the strain gauge directly with your fingers. The oils on your skin will prevent the strain gauge from bonding.

- Stick the strain gauge onto the glass plate with tape.



1.3. Preparing the surface

Initial Cleaning

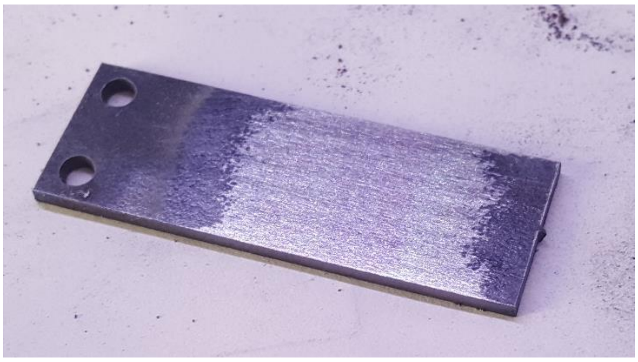
- Roughly mark out the required location of the strain gauge.
- Use the Dremel and 80 grit flapper wheel to remove any paint, excess dirt, rust etc from the surface until the actual material is clearly visible. This area should be about 50 x 50 mm.
- Take care not to remove to much of the material surface creating a change in the geometry or structural integrity of the component.

Cleaning

- Clean the surface with RMS 1 to remove any grease or oil prior to sanding

Sanding the surface

- This is done to remove all surface flaws as well as to give the surface some roughness.
- The exact grain is normally specified by the adhesive manufacturer. (Adhesive Z70 – P220)
- Always sand in circular motion
- Ensure that a large enough area is sanded.



Cleaning the surface

- Clean the surface with RSM1 to remove all dirt caused by the sanding

Marking

- Mark the exact position for the strain gauge
- Use an empty ball point pen or pencil for this

  NB: Never use a sharp scribe.

  NB: Be careful not to create stress concentrations!

Final Cleaning

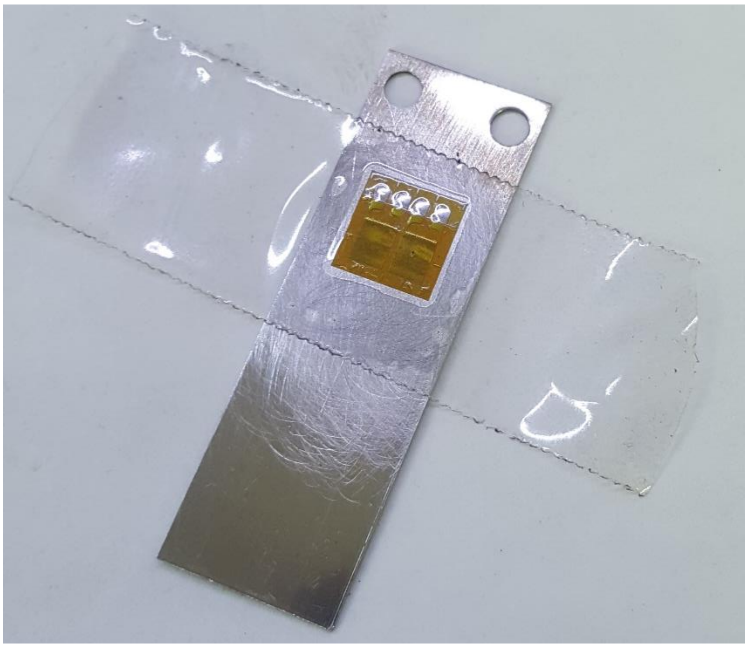
- Clean the area with RMS 1 wiping only in one direction
- Repeat until cleaning pad comes off clean

- Never blow or breath directly on the surface

1.4. Bonding the strain gauge

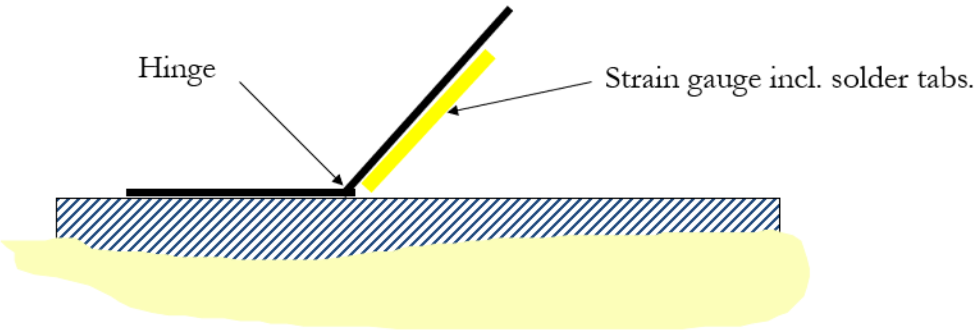
Placing the strain gauge

- Remove the strain gauge from the glass plate with the tape
- Always use tweezers to handle the strain gauge
- Position the strain gauge on the markings using the tape



Lifting the gauge

- Lift/Rotate the strain gauge using the tape as a hinge

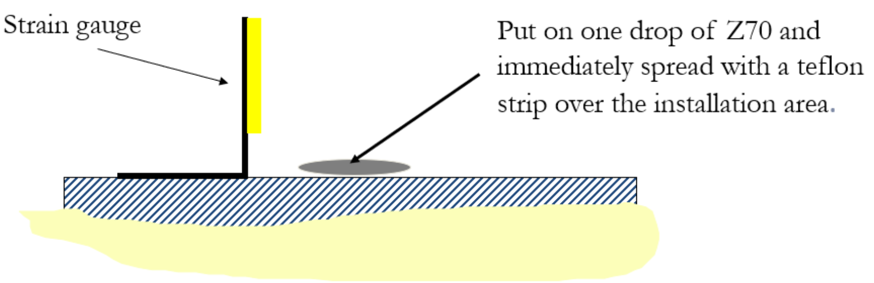


Bonding

- Carefully read the requirements for your adhesive before applying it to the strain gauge. Take note of the following:

- pot life,
- curing time,
- curing temperature,
- contact pressure

- Z70 will be used for this example. Z70 cures very fast so ensure that you have everything ready at this point
- Place a drop of Z70 on bonding area and spread with a piece of Teflon strip



Bonding

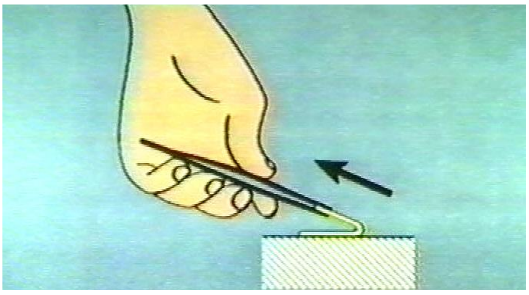
- Put down the strain gauge
- Cover with Teflon foil
- Do this as quickly as possible
- Press down with thumb for 1 minute

  NB: Once you have started applying pressure do not lift your finger.

Removing the tape

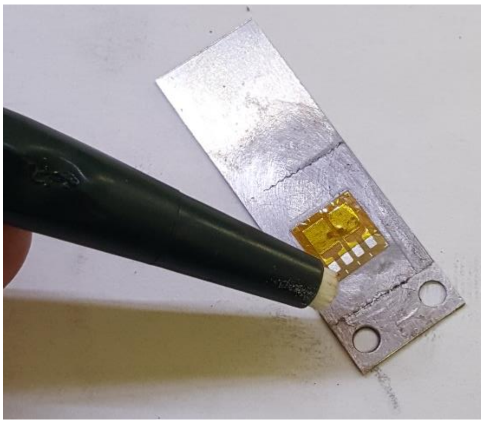
- Z70 needs about 10 min to fully cure before removing the tape

- Remove the tape with tweezers by pulling it off very flat (parallel to the strain gauge)



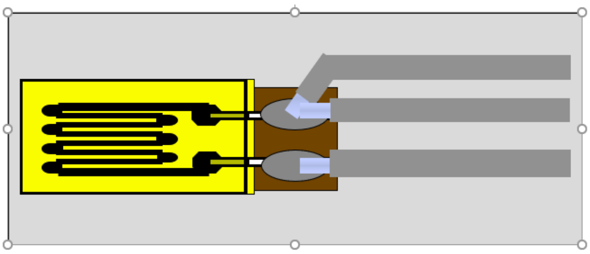
1.5. Soldering

- Clean the solder pads gently with a fibre tip pen
- Add some flux using a flux pen



1.6. Add solder to the solder pads

- Tin the wires by add solder to the wire ends
- Solder the wire to the solder pads without adding solder
- Ensure that the wire insulation overlaps the strain gauge
- Clean the solder tabs with RMS one to remove excess flux
- Rasp away excess flux



1.7. Testing

Visual Testing

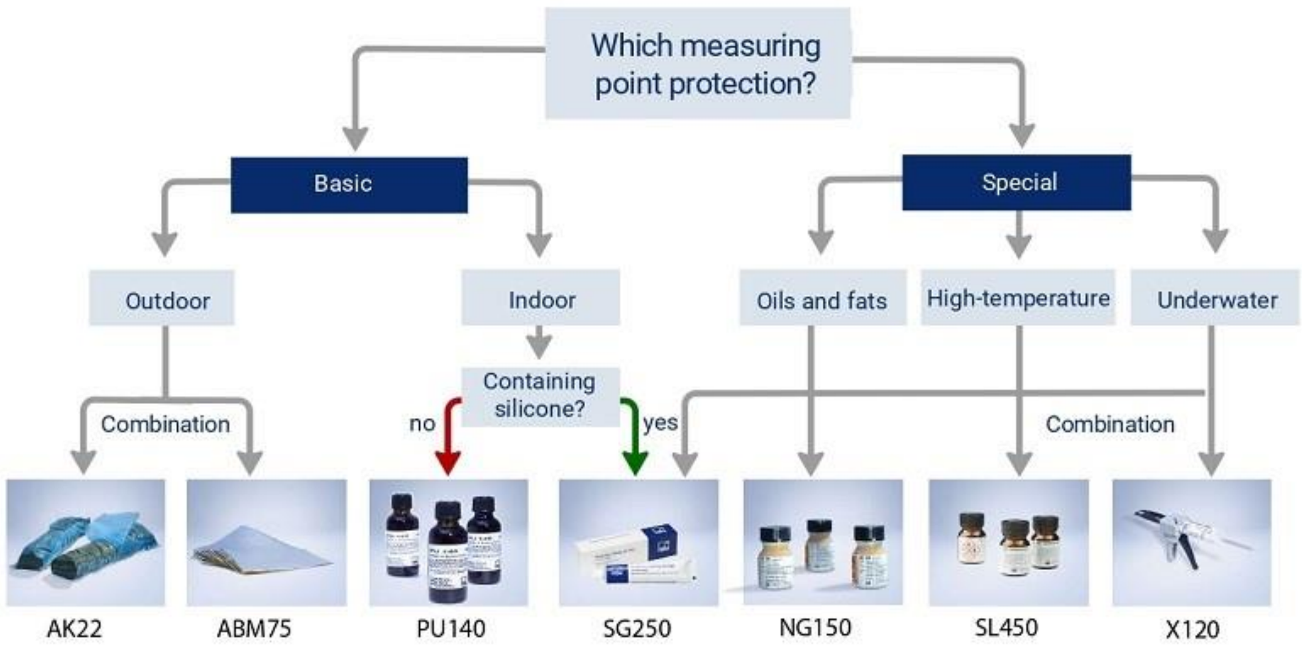
- Air particles under the strain gauge – Redo
- Insufficient adhesion at the edges - Redo
- Poor soldering quality - Redo
- Strain Gauge alignment - Redo

Insulation Testing

- Measure the continuity between the solder pads and material surface. there should be no continuity.
- Measure the resistance at the ends of the connection leads. The resistance value should be within 1 to 2 ohms of the know strain gauge resistance, depending on the accuracy of your multi meter and the cable length.

1.8. Covering/Protection

- Use the flow chart below to select the correct protective covering. A combination of coverings will provide good moister and mechanical protection.



- Cover all soldered connection points and open ends of the cables with the protective agent.

- Bring the cable out from the bottom of the protective covering to prevent moisture collecting on the cable and running into the measurement point.
- Place a strain relief loop on the cable using a self-adhesive cable tie saddle to secure the cable close to the strain gauge.

1.9. Tips

- Wash your hands before strain gauge installation
- Ensure that you have everything you need before hand
- Avoid distractions
- Be as comfortable as possible
- Have sufficient lighting
- Do not rush
- Find ways that work best for you